

# Assignment 1

## 1

(True or False. Provide Explanation) A primary advantage of fixed-length instructions is that they simplify pipelining, since the address of the next instruction can be determined while the current instruction is being decoded.

✓ Answer ✓

True, fixed length instructions very easily allow you to look up the next instruction simply by adding size to your current pointer, allowing for pipelining or parallel execution to happen.

## 2

(True or False. Provide Explanation) IEEE 754 even has a symbol for the result of invalid operations, such as  $0/0$  or subtracting infinity from infinity. This symbol is NaN, for Not a Number. NaN has no usage in the programming and only causes error and should be avoided.

✓ Answer

False, NaN is useful for writing resilient code that can take into account errors and bad computations. While yes, you should typically avoid it if you mean to perform useful computation, it is a good indication a calculation went wrong when it does appear.

## 3

Represent the following numbers using a binary single-precision floating-point number. Your answer should be 32 bit long, ordered by 1 sign bit, 8 exponent bits, and fraction bits:

a

-0.85

✓ Answer

$\text{sign}(-0.85) = -1$   
 $\log_2(0.85) = -0.23 \rightarrow e = -1 \rightarrow 126 \rightarrow 01111110$   
 $\frac{0.85}{2^{-1}} = 1.7 \rightarrow 10110011001100110011001$   
 $-0.85 \rightarrow 1|01111110|10110011001100110011001$

b

1.1

✓ Answer

$$\text{sign}(1.1) = 1$$

$$\log_2(1.1) = 0.14 \rightarrow e = 0 \rightarrow 127 \rightarrow 01111111$$

$$\frac{1.1}{2^0} = 1.1 \rightarrow 00011001100110011001100$$

$$1.1 \rightarrow 0|01111111|00011001100110011001100$$

c

0.75

✓ Answer

$$\text{sign}(0.75) = 1$$

$$\log_2(0.75) = -0.42 \rightarrow e = -1 \rightarrow 126 \rightarrow 01111110$$

$$\frac{0.75}{2^{-1}} = 1.5 \rightarrow 100000000000000000000000$$

$$0.75 \rightarrow 0|01111110|100000000000000000000000$$

4

Write down the binary representation of the decimal number 63.25, assuming

a

IEEE 754 single precision format.

✓ Answer

$$\text{sign}(63.25) = 1$$

$$\log_2(63.25) = 5.98 \rightarrow e = 5 \rightarrow 132 \rightarrow 10000100$$

$$\frac{63.25}{2^5} = 1.9765625 \rightarrow 111110100000000000000000$$

$$63.25 \rightarrow 0|10000100|111110100000000000000000$$

b

IEEE 754 double precision format.

✓ Answer

$$\text{sign}(63.25) = 1$$

$$\log_2(63.25) = 5.98 \rightarrow e = 5 \rightarrow 1028 \rightarrow 10000000100$$

$$\frac{63.25}{2^5} = 1.9765625 \rightarrow 1111101000000000000000000000000000000000000000000000000$$

$$63.25 \rightarrow 0|10000000100|11111010000000000000000000000000000000000000000000000000$$

5

Using a 6-bit normalized binary floating-point format, perform the following addition:  $0.6875 + (-0.171875)$ . Show all steps.

### ✓ Answer

$$\text{sign}(0.6875) = 1$$

$$\log_2(0.6875) = -0.54 \rightarrow e = -1$$

$$\frac{0.6875}{2^{-1}} = 1.375$$

$$0.6875 \rightarrow 1.375(2)^{-1}$$

$$\text{sign}(-0.171875) = -1$$

$$\log_2(0.171875) = -2.54 \rightarrow e = -3$$

$$\frac{0.171875}{2^{-3}} = 1.375$$

$$-0.171875 \rightarrow -1.375(2)^{-3}$$

$$1.375(2)^{-1} + -1.375(2)^{-3}$$

$$= 1.011_2(10_2)^{-1} - 1.011_2(10_2)^{-3}$$

$$= 1.011_2(10_2)^{-1} - 0.01011_2(10_2)^{-1}$$

$$= 1.00001_2(10_2)^{-1}$$

Rounded, 0|010|00

## 6

Using a 6-bit normalized binary floating-point format, perform the following multiplication:  $(-0.75) \times (0.3125)$ . Show all steps.

$$\text{sign}(-0.75) = -1$$

$$\log_2(0.75) = -0.42 \rightarrow e = -1$$

$$\frac{0.75}{2^{-1}} = 1.5 = 1.1_2$$

$$\text{sign}(0.3125) = 1$$

$$\log_2(0.3125) = -1.68 \rightarrow e = -2$$

$$\frac{0.3125}{2^{-2}} = 1.25 = 1.01_2$$

$$e = -1 - 2 = -3$$

$$m = 1.1_2 * 1.01_2 = 1.111_2$$

$$s = -1 * 1 = -1$$

Rounded, 1|000|11, which is denormalized.

The closest normalized number would be 1|001|00

## 7

(A or B? Explain.) Which of the following is correct for a load instruction? Refer to the following figure.

- A. MemtoReg should be set to cause the data from memory to be sent to the register file;
- B. MemtoReg should be set to cause the correct register destination to be sent to the register file.

### ✓ Answer

If `MemtoReg` is 1, we can see that it reads from memory. If it were 0, it would read from the ALU result. So A.

## 8

In this problem, you will extend the multi-cycle datapath as presented in the course. The required figure is provided after the problem. It is possible to decrease the execution time of a program by combining two or more instructions (that would normally take two or more instruction times to execute) into a single instruction that takes only one instruction time (or a fraction more than one) to execute. These hybrid instructions are chosen because the instructions they replace appear often as a group in programs, and they can be implemented without extensive modification to the datapath. Modify the datapath as necessary to allow the implementation of each instruction. Be sure to include complete information about any new signals that appear in the modified datapath. After the datapath is modified, fill in the list of control signals that must appear at each clock cycle in order to implement the instruction. To reduce the number of signals that you have to write, assume that the initial value of all signals is zero. Remember that if you assert a signal, you must de-assert it when necessary. The number of clock cycles needed for each instruction may vary. Use as many as you need, but make the implementation as efficient as possible. The signals for the first clock cycle already appear in the table. You should not have to modify them.

Note: In the figure, ALU stands for arithmetic logic unit, MDR for memory data register, PC for program counter, and IR for instruction register. On the next page, modify the datapath (and control signals, as necessary), and fill in the list of control signals to implement an LDINC Rs, offset (Rt) instruction. The LDINC Rs, offset (Rt) instruction loads the word addressed by the sum of the sign-extended offset (bits 15-0) and the value in Rt (bits 25-21) into register Rs (bits 20-16). The value in register Rt is also auto-incremented by 4 so that the next execution of the instruction will load the next word in memory.

### ✓ Answer

| Clock Cycle | Functional Description                  | Signals                                                                                            |
|-------------|-----------------------------------------|----------------------------------------------------------------------------------------------------|
| 1           | Instruction → IR<br>$PC = PC + 4$       | MemRead=1<br>ALUSrcA=0<br>lrd=0<br>IRwrite=1<br>ALUSrcB=01<br>ALUOp=00<br>PCWrite=1<br>PCSource=00 |
| 2           | Wait for register values to be loaded   | *=0                                                                                                |
| 3           | $ALU = \text{offset}(15-0) + Rt(25-21)$ | *=0<br>ALUSrcA=1<br>ALUSrcB=10<br>ALUOp=00                                                         |

| Clock Cycle | Functional Description             | Signals                                                                            |
|-------------|------------------------------------|------------------------------------------------------------------------------------|
| 4           | Load ALU address into MDR          | *=0<br>lorD=1<br>MemRead=1                                                         |
| 5           | Write MDR to Rs<br>$ALU = R_t + 4$ | *=0<br>MemtoReg=1<br>RegDst=0<br>RegWrite=1<br>ALUSrcA=1<br>ALUSrcB=01<br>ALUOp=00 |
| 6           | $R_t = ALU$                        | *=0<br>RegWrite=1<br>RegDst=1<br>MemtoReg=0                                        |

## 9

When processor designers consider a possible improvement to the processor datapath, the decision usually depends on the cost/performance trade-off. In the following two problems, assume that we are starting with a datapath from the following figure, where I-Mem, Add, Mux, ALU, Regs, DMem, and Control blocks have latencies of 370 ps, 100 ps, 30 ps, 120 ps, 200 ps, 350 ps, and 100 ps, respectively. Consider the addition of a multiplier to the ALU. This addition will add 300 ps to the latency of the ALU. The result will be 5% fewer instructions executed since we will no longer need to emulate the MUL instruction

| Path    | Latency (ps) |
|---------|--------------|
| I-Mem   | 370          |
| Add     | 100          |
| Mux     | 30           |
| ALU     | 120          |
| Regs    | 200          |
| DMem    | 350          |
| Control | 350          |

## a

What is the clock cycle time with and without this improvement?

### ✓ Answer

The longest cycle would be I-Mem (and Control), Regs, Mux, ALU, DMem, Mux

= 1100 ps  
Trevor Nichols

Since the ALU is a time-blocking feature of this cycle, the new clock cycle will be 1400 ps, which results in a 21.43% slower clock.

**b**

What is the speedup achieved by adding this improvement

✓ **Answer**

Since the clock is 5% fewer instructions, our total speed change will be 15.357% slower.